João Ferri

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Education

PhD **University of Sao Paulo**, Physics – Sao Paulo, Brazil 2022 – present

MSc **University of Sao Paulo**, Physics – Sao Paulo, Brazil 2020 – 2022

BSc **University of Sao Paulo**, Physics – Sao Paulo, Brazil 2016 – 2019

Research Experience

PhD candidate in physics, University of Sao Paulo – Brazil May 2022 – present
Advisor - Luis Raul Weber Abramo – Grant: CNPq 140344/2022-5

- Multi-tracer formalism
- Fourier/Harmonic/Real space equivalences
- Cross-correlation of galaxies and gravitational wave events

Guest researcher, Kavli IPMU – Tokyo, Japan Oct 2024 – Apr 2025
Advisor - Elisa Ferreira Mauricio Gouvea – Grant: CAPES 88881.982198/2024-01

- Consistency analysis of Hyper-Suprime-Camera galaxy shear data

Master degree in Physics, University of Sao Paulo – Brazil Feb 2020 – Apr 2022
Advisor - Luis Raul Weber Abramo – Grant: CNPq 132397/2020-0

- Multi-tracer formalism in harmonic space
- Full-sky redshift-space distortions in harmonic space

Undergrad research, University of Sao Paulo – Brazil Jan 2019 – Dec 2019
Advisor - Luis Raul Weber Abramo

- Full-sky redshift-space distortions in harmonic space

Research Interests

Multi-tracer formalism
Collaborators: Luis Raul Weber Abramo, Ian Tashiro, Arthur Loureiro

- Analytical computations of multi-tracer power spectrum covariance matrices accounting for redshift-space distortions
- Tracer number optimization for galaxy surveys

Gravitational waves
Collaborators: Miguel Quartin, Riccardo Sturani, Tessa Baker, Charles Dalang, Luis Raul Weber Abramo, Ian Tashiro, Isabella Matos, Bernardo Veronese

- Using the angular cross-correlation between GW and galaxy maps to constrain the expansion history of the late-Universe
- Investigate the correspondence between line-of-sight and spatial correlation methods to constrain GW physics

Galaxy shear
Collaborators: Elisa Ferreira Mauricio Gouvea, Masahiro Takada, Ryo Terasawa

- Consistency analysis of Hyper-Suprime-Camera (HSC) Y3 galaxy shear data: comparing analyses done in real and harmonic space; employing emulators for small-scale modeling

Awards

Best poster award Nov 2024

Kashiwa-no-ha Dark Matter and Cosmology Symposium – Kavli IPMU, Tokyo, Japan

- "*A Hubble diagram from the cross-correlations of galaxies and dark sirens*"

Publications

"First measurement of the Hubble constant from gravitational wave-galaxy cross-correlations" Nov 2025

Isabela Matos, Charles Dalang, Tessa Baker, Raul Abramo, ***João Ferri***, Miguel Quartin

In preparation for PRL. In ArXiv by November

"The contribution from smalls scales on two-point shear analysis: comparison between power spectrum and correlation function" Oct 2025

Ferri, J., *Ferreira, E.*

In preparation for JCAP. In arXiv by October

"A robust cosmic standard ruler from the cross-correlations of galaxies and dark sirens" Apr 2025

Ferri, J., *Tashiro, I.L.*, *Abramo, L.R.*, *Matos, I.*, *Quartin, M.*, *Sturani, R.*

10.1088/1475-7516/2025/04/008 (Journal of Cosmology and Astroparticle Physics)

"Fisher matrix for the angular power spectrum of multi-tracer galaxy surveys" Aug 2022

Abramo, L.R., ***Ferri, J.***, *Tashiro, I. L.*, *Loureiro, A.*

10.1088/1475-7516/2022/08/073 (Journal of Cosmology and Astroparticle Physics)

"Fisher matrix for multiple tracers: the information in the cross-spectra" Apr 2022

Abramo, L.R, ***Ferri, J.*** , *Tashiro, I.L.*

10.1088/1475-7516/2022/04/013 (Journal of Cosmology and Astroparticle Physics)

Talks

A robust cosmic standard ruler from the cross-correlations of galaxies and dark sirens

- Kavli IPMU, Astro Lunch Seminars, Jan 2025 (in person)
- Chiba University, Cosmology Group Meeting, Dec 2024 (in person)
- ICTP-SAIFR, São Paulo Research Group meetings Astro & Cosmo, Aug 2024 (in person)

What can we gain from small scales in shear analysis?

- ICTP-SAIFR, São Paulo Research Group meetings Astro & Cosmo, Aug 2025 (in person)

Skills & Interests

Programming: Python, Latex, C

Tools: Linux, HEALPix, NaMaster, GLASS, CosmoSIS, CCL, Jupyter/VS Code

Languages: English (proficient, DET 135/160), Japanese (beginner, JLPT N3), Portuguese (native speaker)

Hobbies: Cars, Music

Teaching & Other Activities

Teaching Assistant, University of Sao Paulo Mar 2022 – Aug 2022

Electromagnetism Lab

Teaching Assistant, University of Sao Paulo Aug 2023 – Dec 2023

Classical Mechanics II

Internship Assistant, Kavli IPMU — Apr 2025

Introduction to Fisher Matrix formalism, Cluster & Jupyter Lab usage

References

Dr. Luis Raul Weber Abramo: abramo@if.usp.br

Dr. Elisa Gouvea Mauricio Ferreira: elisa.ferreira@ipmu.jp

Dr. Riccardo Sturani: riccardo.sturani@unesp.br

Dr. Miguel Quartin : miguel.q.fisica@gmail.com

Dr. Isabela Santiago De Matos: isabela.matos@port.ac.uk